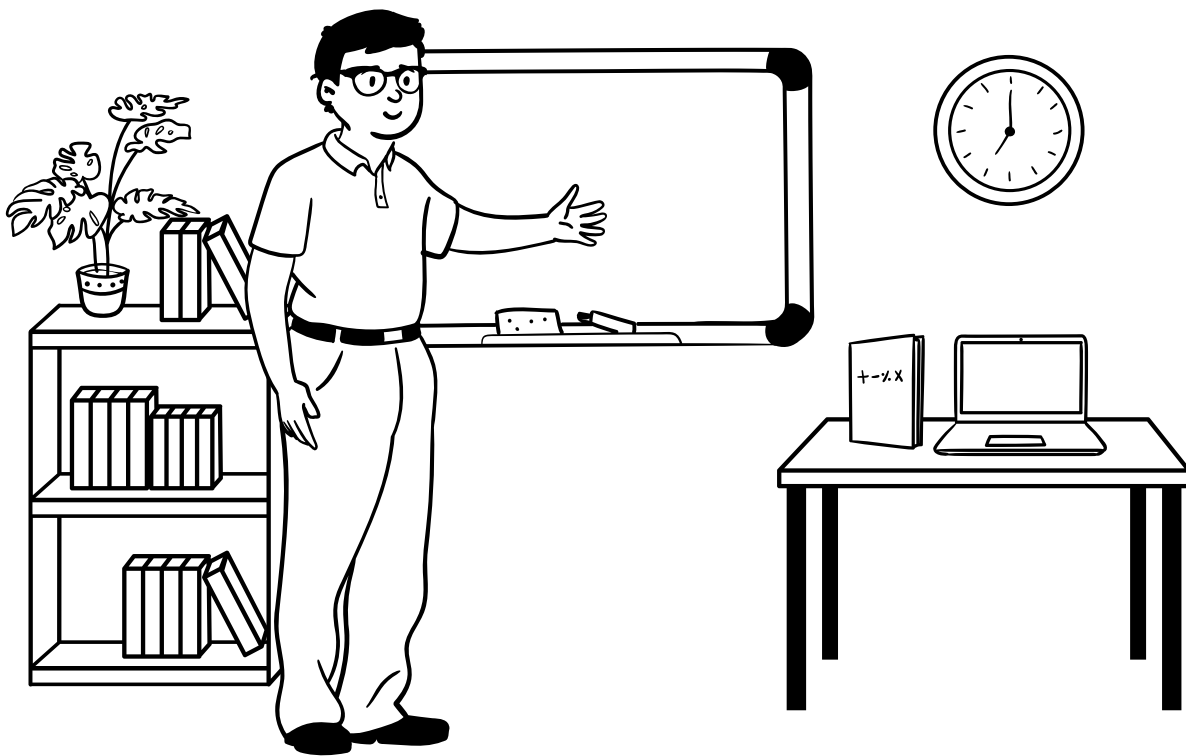


MAT 380 Spring 2026

Real Analysis

Packet 1:

The Real Numbers and Convergence of Sequences



Name: _____

Contents

Syllabus and Course Policies	1
Introduction to the Real Numbers	7
Wednesday, February 4 - Why Real Numbers? Thinking Like an Analyst	7
Friday, February 6 - The Axiom of Completeness	9
Monday, February 9 - $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{N}$: Density and Cardinality	14
Wednesday, February 11 - Proofs Review and Workshop	19
Convergence of Sequences	24
Friday, February 14 - Convergence of Sequences: Intuition	24
Monday, February 16 - Convergence of Sequences: Formal Definition	27
Wednesday, February 18 - The Monotone Convergence Theorem	30
Friday, February 20 - Algebraic Limit Theorem	34
Monday, February 23 - Subsequences and More Convergence Theorems	36
Wednesday, February 25 - The Bolzano-Weierstrass Theorem	39
Friday, February 27 - Cauchy Sequences	43
Monday, March 2 - The Cauchy Criterion	44
Wednesday, March 4 - Review: The Five Characterizations of Completeness	47

MAT 380 – Real Analysis

Spring 2026

Instructor: Dr. Dana Neidmann

Pronouns: he/him

Pronunciation: **Nyed**-mun

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Office Hours:

Monday 3-4pm

Tuesday 10-11am

Wednesday 3-4pm

Thursday 2-4pm

Friday 10:15-11:15am

Or by appointment on Navigate



Class Information

Room: Olin 122

Time: 9:10-10:10am MWF

Textbooks:

Understanding Analysis, 2nd Edition by Stephen Abbott (for reference and practice problems)

How to Think About Analysis by Lara Alcock (for readings and reflection)

Course Description

(from the course catalog) A study of the theoretical foundations of single variable calculus with careful attention to definition and proof. Topics include properties of real numbers, limits, continuity, differentiation, and integration.

Course Assessment

This course is meant to teach you not only an understanding of topics in Real Analysis, but how to tackle problems, craft proofs, and communicate effectively. Daily engagement will be assessed via the *Analysis Journal*, and your problem-solving, proof-writing, and communication skills will be assessed via *Pick & Prove*, *Present*, and *Perform*.

Grade Item	Grade %
Analysis Journal (Preparation and Attendance)	10%
Pick & Prove (Weekly Homework)	40%
Present (2+ presentations throughout semester + final)	5%
Perform (3 midterm exams)	10% each
Final Project	15%

Weights of each of these categories are above, and tentatively letter grade cutoffs will follow a standard scale: A (93-100), A- (90-92), B+ (87-89), B (83-86), etc.

Analysis Journal (Daily Engagement, Preparation, and Attendance)

The daily work in this course will be keeping an Analysis Journal that will get you thinking about what we have covered or about the content to come. Check the handout available on Moodle for more information about what the Analysis Journal is all about and what should be included in it.

In class, I will visually check that you've done the journal writing for that day. Since I will check it in class, this will also function as an attendance grade. If you miss class for an excused absence, I may allow you to turn in your journal remotely. The two lowest grades will be dropped.

Pick & Prove, Present, and Perform

You will learn proof techniques, but I also want you to improve throughout the semester. Thus, I have devised the following main assessment scheme:

Pick & Prove (Weekly Homework)

Each weekly homework assignment will consist of a selection of problems related to the material, in 3 difficulty/length categories: ★, ★★, and ★★★. For each homework assignment, you will **pick** a selection of problems that add up to six stars.

★: More straightforward concept and definition application, often not involving rigorous proof.

★★: Basic proofs, using straightforward techniques or that are similar to ones we did in class.

★★★: More involved proofs, which require some degree of creativity or insight to effectively prove.

So, you can do fewer problems and pick two ★★★ problems for a bit of a challenge, or if you still need solidifying on the basic material, you can do four ★ problems and a ★★ problem. *You must complete at least one ★★ or ★★★ problem per assignment.* I suggest you give each problem a good try (see "Perform" below as to why this might be a good idea) and pick the ones you feel most confident about in your solutions.

Each homework will be graded out of 12 points: ★ problems are graded out of 2, ★★ problems are graded out of 4, and ★★★ problems are graded out of 6. Problems will be graded on both accuracy and style.

On each homework assignment, you can earn two bonus points for typesetting your solutions in LaTeX. If you choose to do so, please print the homework assignment before turning it in.

Present (2+ homework presentations per semester)

Homework will be due in class on Friday, and each Friday I'll randomly select two or three students to come present one of their proofs on the board. You won't be graded on your accuracy, but on your ability to clearly explain what you did. See the rubric on Moodle for more information about how you will be assessed.

Perform (3 Midterm Exams)

You'll notice that all three midterm exams are worth less in total than your weekly homework. This is intentional. I am more concerned with your ability to creatively determine solutions and craft proofs when you have time to visit, revisit, and collaborate with your peers than being able to perform on a test. However, I need to assess *your* understanding independently from your peers.

Thus, to promote a growth-based mindset, *all* the midterm problems will be either proofs we went over in class, or problems from your weekly homework assignments. This means that even if you chose only ★ and ★★ weekly homework questions, you still need to eventually figure out how to solve the ★★★ problems.

Tentative dates of Midterm Exams:

- **Friday, March 6:** Real Numbers and Sequences
- **Friday, April 10:** Topology and Continuity
- **Friday, May 8:** Differentiability and Integrability

Final Project

Your goal for this project will be to explore an analysis topic of your choice beyond what we have done in class and teach it to your fellow classmates. You may work alone or in pairs.

During our final exam slot on **Monday, May 18 from 8:30-11:30am**, you will give a 20-25-minute Beamer (LaTeX) presentation on your topic and turn in a brief overview document of the relevant theorems, exposition, and examples with citations. More instructions and rubrics will be forthcoming.

Getting Help

Office Hours

Please come to office hours! Having extra time together outside of class is invaluable. Some things you can ask about in office hours include:

- I'm having trouble with this specific homework problem. Here's what I've tried so far.
- Can we review this concept from class? I didn't really get it.
- I know this was a long time ago, but I still don't get this concept from a few weeks ago. Can we review it?
- I hear you saying some fact that we're just assuming, but I don't understand why it's true. Can you explain it?
- I saw something interesting mentioned in the book that you didn't talk about, can we discuss it?
- I don't really know how to study for the exams, can you help?
- I'm feeling really stressed, can we work out a plan so I can still succeed in this class?

1-on-1 Appointments

If you can't make office hours, or if you have a question of a more personal nature or something that would be hard to go over in my office hours, I would love to meet with you for a short meeting. You can set these up using the Navigate availability link in the QR code at the top of the syllabus (also in my email signature). I prefer this to asking in person, because I am very forgetful! These Navigate appointments are scheduled on a first-come, first-served basis.

Email

You can certainly email me if you have questions. Often trying to answer complex mathematical questions is difficult over email, so I encourage you to meet me in person for these types of questions. However, email is the perfect avenue for more administrative questions or quick clarifications. I will try to respond to any email in a timely manner (within 24 hours during the week).

Accommodations

I encourage students with disabilities such as chronic medical conditions, learning disabilities, and mental health concerns as well as students dealing with exceptional circumstances to seek official accommodations. However, if you don't have documented reasons, but feel that I can be accommodating to you in a specific way, please come and talk to me and we can have a conversation about how I can best address your needs.

Staying Healthy

Please do what you can to keep yourself and your classmates healthy. If you feel ill, please go to Parsons, do not come to class, and try to get the notes from someone. I can answer questions you have about the topics you missed but it is not feasible for me to reteach entire classes. If you end up missing a lot of class, please contact me and we can discuss paths for making up a great deal of work.

Mental health is also a part of staying healthy. Please take care of yourself. If you find that your mental health is interfering with your ability to succeed in this course, come talk to me sooner rather than later, and we can perhaps talk through a plan to ensure your success. I can also help point you to campus resources.

Course Policies

This is where the syllabus stops being fun and starts getting rather litigious and lengthy. However, you are still responsible for knowing these policies, and “I didn’t know that was how it worked” is not a valid excuse.

Expectations for Classroom Behavior

I expect everyone to bring their full selves to class. This means engaging in group discussion activities, answering questions, asking questions to get us back on track if you get lost, and taking notes. I expect you to stay focused and not be on your phone. If this becomes a problem, I may have to start collecting phones at the start of class.

Above all, please be kind to one another. Be kind to your fellow students and to me, and I will do my best to be kind to you. Assume the best in people. We are all learners, and we all need to feel safe in this collaborative space we are creating. Please let me know if there is anything I can do to improve your classroom environment.

Absence

I will drop the two lowest Analysis Journal grades, which means that you can have up to two unexcused absences without penalty. I won’t ask you to give me a reason why you miss class on these days, and these two absences are perfect for use for personal events or mental health days. If you need to miss class for an excusable reason (sickness or college-sponsored event), I may excuse your absence and instead have you turn in your Analysis Journal remotely. However, an excused absence must be negotiated before class unless in an extreme emergency. Missing an exam requires documentation.

Late Work

Late work will not be accepted. If you are absent on the day the homework is due, please make sure it gets to me on time (preferably delivered by a friend/classmate). Absence does not excuse you from deadlines.

Occasionally, and in exceptional circumstances, I may grant an extension to make sure that your work is the best that you can present. These circumstances do *not* include “I had a lot of work in other classes.” In any case, extensions *must* be negotiated before the assignment is due.

Community, Collaboration, and Academic Honesty

I encourage you to find old or new friends in the class, and I encourage you to work on the homework together. Even though I encourage collaboration, I do expect your work to be your own. Make sure if you’re sharing ideas, there’s an accompanying conversation, and you’re not just looking at someone’s paper or computer screen and copying what they did. All written work should be composed by you, even if you discuss ideas with others. On the homework, ***you must acknowledge the sources you consulted***: i.e. other students, me, your mom, etc..

Here are some examples of the difference between collaborating and academic dishonesty:

Accepted Collaboration	Academic Dishonesty
“Can you explain your proof to me?”	“Can I see your paper to read your proof?”
“Thanks for sharing your approach! Let’s write our solutions separately and then trade to check over each other’s work.”	“You had a really good idea, can I use what you wrote?”
“Let’s look in the textbook to see if there’s a similar problem worked out that we can use to figure out this one.”	“Let’s search online to see if someone’s discussed this problem on StackExchange or Chegg.”
“Can you read over my proof and let me know if anything could be improved?”	“I’m going to ask ChatGPT to proofread my solution.”

Technology Use

You should be filling out the guided notes every day, but you are welcome to do that on a tablet rather than on paper if you prefer. This course does not lend itself well to taking notes via typing, unless you’re really good at LaTeX. No calculators will be needed or allowed on assessments.

Real Analysis is not an appropriate place to use ChatGPT or any other GenAI in *any* capacity. Mathematical ideas should come from your own brain or conversations with real people. In addition, for Real Analysis ChatGPT will just *lie* about important concepts but will make it sound very convincing. Please do not use it at all on your homework or final projects. This includes for brainstorming, editing, revising, and proof writing – these are also skills that I want *you* to develop.

The only approved use of GenAI or online searching is in helping typeset things into LaTeX. You may ask about how to typeset certain things (e.g. “How do I make the real numbers symbol” or “how to make a list in LaTeX”), but you should not copy your own work into ChatGPT for it to format for you.

Tentative Schedule

February 4-11: Introduction to the Real Numbers (UA Chapter 1)

- Wednesday, February 11: Proof Review and Workshop
- Friday, February 13: HW#1 Due (First Problem Presentation)

February 13-March 6: Convergence of Sequences (UA Chapter 2)

- Friday, February 20: HW#2 Due
- Friday, February 27: HW#3 Due
- Friday, March 6: Midterm Exam #1

March 9-20: Topology of the Real Numbers (UA Chapter 3)

- Friday, March 13: HW#4 Due
- Friday, March 20: HW#5 Due

March 30-April 10: Functions and Continuity (UA Chapter 4)

- Friday, April 3: HW#6 Due
- Friday, April 10: Midterm Exam #2

April 13-April 24: Differentiability (UA Chapter 5)

- Friday, April 17: HW#7 Due
- Friday, April 24: HW#8 Due

April 27-May 11: Integrability (UA Chapter 7)

- Friday, May 1: HW#9 Due
- Friday, May 8: Midterm Exam #3

The instructor maintains the right to make changes in the syllabus when necessary. Any changes made will be announced during class, and the date of the update will be posted to Moodle.

Understanding Analysis Sections Covered: Sections 1.2 and 8.2 (definition of metric only)

Definitions and Theorems to include in Analysis Journal:

- Definition 8.2.1 (metric, metric space)
-

Theorem 1.2.6

$$\forall a, b \in \mathbb{R}, a = b \iff \forall \epsilon > 0, |a - b| < \epsilon.$$

1. (a) Translate the theorem into words and write its translation in the box above. Are there any symbols you aren't familiar with?
(b) Justify the statement to yourself. What would we need to do to prove this statement?
(c) Carefully prove the theorem.

- (d) In our proof, where did we use the fact that $a, b \in \mathbb{R}$? What *assumptions* about the real numbers are necessary here?

Definition 8.2.1: Metric Space

A **metric space** is a set X along with a function $d : X \times X \rightarrow \mathbb{R}$ such that for all $x, y \in X$,

- (i)
- (ii)
- (iii) **Triangle Inequality:**

The function d is called a **metric** on X .

After Class:**Pick & Prove Problems:****★ Exercise 8.2.1**

Decide which of the following are metrics on $X = \mathbb{R}^2$. For each, we let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be points in the plane. If they are metrics, use the definition of metric to confirm. If they are not, illustrate why they are not.

- (a) $d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$
- (b) $d(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$
- (c) $d(x, y) = |x_1 x_2 + y_1 y_2|$

★ Exercise 1.2.10

Decide which of the following statements are true statements. Provide a short justification for those that are valid and a counterexample for those that are not.

- (a) Two real numbers satisfy $a < b$ if and only if $a < b + \epsilon$ for every $\epsilon > 0$.
- (b) Two real numbers satisfy $a < b$ if $a < b + \epsilon$ for every $\epsilon > 0$.
- (c) Two real numbers satisfy $a \leq b$ if and only if $a < b + \epsilon$ for every $\epsilon > 0$.

★ ★ ★ Exercise 1.2.6

Triangle Inequality: For all real numbers a and b , $|a + b| \leq |a| + |b|$.

- (a) Give a short proof of the triangle inequality in the special case where a and b have the same sign.
- (b) Find an efficient proof for all cases of the triangle inequality at once by first demonstrating that $(a + b)^2 \leq (|a| + |b|)^2$.
- (c) Prove using the triangle inequality that $|a - b| \leq |a - c| + |c - d| + |d - b|$ for all a, b, c , and d . (It may help to review the logic on p.8 of UA where they show that $|a - b| \leq |a - c| + |c - b|$ using the triangle inequality.)
- (d) Prove $||a| - |b|| \leq |a - b|$. (The unremarkable identity $a = a - b + b$ may be useful.)
- (e) Given the metric $d(a, b) = |a - b|$ on the real numbers, write a statement that is the “triangle inequality” for the metric (it’s not the triangle inequality above).

Understanding Analysis Sections Covered: Sections 1.3-1.4

Definitions and Theorems to include in Analysis Journal:

- Axiom of Completeness
- Definition 1.3.1 (upper bound, lower bound)
- Definition 1.3.2 (supremum, infimum)
- Definition 1.3.4 (maximum, minimum)
- Lemma 1.3.8
- Theorem 1.4.1 (Nested Interval Property)
- Cut Property (Exercise 1.3.10)

Before Class:

Readings: *How to Think About Analysis* by Lara Alcock

- Chapter 4: Learning Analysis (pp. 45-52)
- Conclusion (pp.217-222)
- Chapter 10: The Real Numbers, Sections 10.1, 10.4, 10.5 (skip stuff about sequences after middle of p. 213)

Analysis Journal Prompts:

- Write a concrete plan for keeping up with your Analysis work. Which advice from Alcock stands out to you? What have you found useful in the past?
- Come up with a plan if you're getting frustrated with your homework. What kind of self-talk will you employ? What will you actually do?
- What is an axiom, and how is it different from a definition or theorem?
- Start your definition list with the definition of a metric from last class (from here on out, I won't remind you to do this, but please update your definition list after every class).
- Start your "Questions About Analysis" with any questions you may have (from here on out, I won't remind you to do this, but please keep writing questions).

1. Consider the phrase "*there are no holes in the real number line.*" What do you think this means?

Axiom of Completeness

2. *Discuss from Journal:* What is an axiom, and how is it different than a definition or theorem?

Definitions

Upper Bound:

Least Upper Bound (_____):

3. How is a least upper bound different than a *maximum*?
4. Discuss the infima and suprema of the following sets.
- (a) $(0, 2)$ (b) $[0, 2]$
- (c) $A = \{\frac{1}{n} : n \in \mathbb{N}\}$
5. Why is $\mathbb{Q}$ not complete? Construct a set in $\mathbb{Q}$ that does not contain its supremum.

Lemma 1.3.8

Assume $s \in \mathbb{R}$ is an upper bound for $A \subseteq \mathbb{R}$. Then $s = \sup A \iff \forall \epsilon > 0, \exists a \in A$ such that $s - \epsilon < a$.

Proof.

□

Theorem 1.4.1: Nested Interval Property

For each $n \in \mathbb{N}$, assume we are given a closed interval $I_n = [a_n, b_n] \subseteq \mathbb{R}$. Assume for all n , $I_{n+1} \subseteq I_n$.

Proof.

□

6. Find the following intersections:

(a) $\bigcap_{n=1}^{\infty} \left[-\frac{1}{n}, \frac{1}{n} \right]$

(b) $\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n} \right)$

7. (a) Is the nested interval property true for open intervals? (Homework)

(b) Must the intervals be *bounded*?

8. Must a finite set contain its supremum?

9. *True or False:*

(a) If $\sup A \leq \sup B$, $\exists b \in B$ such that b is an upper bound for A .

(b) If $\sup A < \inf B$, then $\exists c \in \mathbb{R}$ such that $a < c < b$ for all $a \in A$ and $b \in B$.

10. Can you find two sets A and B such that

- $A \cap B = \emptyset$
- $\sup A = \sup B$
- $\sup A \notin A$ and $\sup B \notin B$

After Class:

Pick & Prove Problems:

★ **Exercise 1.3.8**

Compute without proofs, the suprema and infima (if they exist) of the following sets:

- (a) $\{m/n : m, n \in \mathbb{N} \text{ with } m < n\}$
- (b) $\{(-1)^m/n : m, n \in \mathbb{N}\}$
- (c) $\{n/(3n+1) : n \in \mathbb{N}\}$
- (d) $\{m/(m+n) : m, n \in \mathbb{N}\}$

★ **Exercise 1.4.3**

Prove that $\bigcap_{n=1}^{\infty} (0, 1/n) = \emptyset$. Notice that this demonstrates that the intervals in the Nested Interval Property must be closed for the conclusion of the theorem to hold.

★★ **Exercise 1.3.1**

- (a) Write a formal definition in the style of 1.3.2 for the infimum or greatest lower bound of a set.
- (b) State and prove a version of Lemma 1.3.8 for greatest lower bounds.

★★ **Exercise 1.3.3**

- (a) Let A be nonempty and bounded below, and define $B = \{b \in \mathbb{R} : b \text{ is a lower bound for } A\}$. Show that $\sup B = \inf A$.
- (b) Use (a) to explain why there is no need to assert that greatest lower bounds exist as part of the Axiom of Completeness.

★★★ **Exercise 1.3.10**

Cut Property: If A and B are nonempty, disjoint sets with $A \cup B = \mathbb{R}$ and $a < b$ for all $a \in A$ and $b \in B$, then there exists $c \in \mathbb{R}$ such that $x \leq c$ whenever $x \in A$ and $x \geq c$ whenever $x \in B$.

- (a) Use the Axiom of Completeness to prove the Cut Property.
- (b) Show that the implication goes the other way; that is, assume $\mathbb{R}$ possesses the Cut Property and let E be a nonempty set that is bounded above. Prove $\sup E$ exists.
- (c) The punchline of parts (a) and (b) is that the Cut Property could be used in place of the Axiom of Completeness as the fundamental axiom that distinguishes the rational numbers from the real numbers. To drive this point home, give a concrete example showing that the Cut Property is not a valid statement when $\mathbb{R}$ is replaced by $\mathbb{Q}$.

Understanding Analysis Sections Covered: Sections 1.4-1.5

Definitions and Theorems to include in Analysis Journal:

- Theorem 1.4.2 (Archimedean Property)
- Theorem 1.4.3 (Density of $\mathbb{Q}$ in $\mathbb{R}$)
- Definition 1.5.1 (one-to-one and onto)
- Definition 1.5.2 (sets have same cardinality)
- Definition 1.5.5 (countable, uncountable)
- Theorems 1.5.6, 1.5.7, 1.5.8
- Schroder-Bernstein Theorem (Exercise 1.5.11)

Before Class:

Readings: None

Analysis Journal Prompts:

- What do you already know (from previous classes) about $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{N}$? Compare and contrast each set (you could make a venn diagram). What about their structure makes each unique? (for this class we take $\mathbb{N} = \{1, 2, 3, 4, \dots\}$.)

Theorem 1.4.2: Archimedean Property

- (i)
- (ii)

Proof.

Theorem 1.4.3: Density of $\mathbb{Q}$ in $\mathbb{R}$

(i)

(ii)

Proof.

□

1. Which of the following sets are dense in $\mathbb{R}$?

(a) $\mathbb{R}$

(b) $\mathbb{Q}$

(c) $\mathbb{N}$

(d) $\mathbb{I}$

Cardinality

We say A has the same **cardinality** as B , written $A \sim B$ if

A is **countable** if

B is **uncountable** if

Countable Sets

Uncountable Sets

Continuum Hypothesis

There is no set whose cardinality is strictly between that of the natural numbers and the real numbers.

Proof sketches:

- $\mathbb{Q}$ is countable.
- $[0, 1]$ is uncountable (**Cantor's Diagonalization Argument**).
- $\mathbb{R}$ is uncountable (using NIP).
- If $A \subseteq B$ and B is countable, then A is either countable or finite.
- If A and B are countable, then $A \cup B$ is countable.

- The finite union of countable sets is countable.
- The countable union of countable sets is countable.
- The set of all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ is uncountable.
- $P(\mathbb{N})$ is uncountable.

Theorem 1.6.2: Cantor's Theorem

Given any set A , there does not exist a function $f : A \rightarrow P(A)$ that is onto.

Proof.

□

After Class:

Pick & Prove Problems:

★ Exercise 1.4.6

Which of the following sets are dense in $\mathbb{R}$? Take $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ in every case. Justify your answers.

- (a) The set of all rational numbers p/q with $q \leq 10$.
- (b) The set of all rational numbers p/q with q a power of 2.
- (c) The set of all rational numbers p/q with $10|p| \geq q$.

★ Exercise 1.5.6

- (a) Give an example of a countable collection of disjoint open intervals.
- (b) Give an example of an uncountable collection of disjoint open intervals, or argue that no such collection exists.

★★ Exercise 1.5.7

Consider the open interval $(0, 1)$ and let S be the set of points in the open unit square; that is $S = \{(x, y) : 0 < x, y < 1\}$.

- (a) Find a 1-1 function that maps $(0, 1)$ into, but not necessarily onto, S . (This is easy.)
- (b) Use the fact that every real number has a decimal expansion to produce a 1-1 function that maps S into $(0, 1)$. (To avoid the ambiguity of whether the decimal expansion is written as .234999... or .235, use the convention to always use the terminating expansion.) Try to turn two decimal expansions into one! Note that I'm not making you discuss whether the function is onto or not.
- (c) Briefly explain why the Schroder-Bernstein Theorem discussed in Exercise 1.5.11 can now be applied to conclude that $(0, 1) \sim S$.

★★★ Exercise 1.5.9

A real number $x \in \mathbb{R}$ is called *algebraic* if it is a root of a polynomial with integer coefficients. Real numbers that are not algebraic are called *transcendental* numbers.

- (a) Show that $\sqrt{2}$, $\sqrt[3]{2}$, and $\sqrt{3} + \sqrt{2}$ are all algebraic.
- (b) Fix $n \in \mathbb{N}$, and let A_n be the algebraic numbers obtained as roots of polynomials with integer coefficients that have degree n . Using the fact that every polynomial has a finite number of roots, show that A_n is countable. *Hint: You will need Theorem 1.5.8.*
- (c) Now, show that the set of all algebraic numbers is countable. What may we then conclude about the set of transcendental numbers?

“It is well known that the man who first made public the theory of irrationals perished in a shipwreck in order that the inexpressible and unimaginable should ever remain veiled. And so the guilty man, who fortuitously touched on and revealed this aspect of living things, was taken to the place where he began and there is forever beaten by the waves.”

– Proclus Diadochus, in a 5th century commentary on Euclid’s Elements

Before Class:

Readings: *How to Think about Analysis*, Chapter 3: Proofs (pp. 31-44)

Analysis Journal Prompts:

- Use the axioms, definitions, and theorems we have learned so far to create a flowchart in the style of Alcock. (Again, throughout the course you should keep up with this and I won’t remind you again.)
- Follow the Self-Explanation strategy in Section 3.5 of Alcock to self-explain the proofs of Theorem 1.1.1 (a proof you have probably seen) and Theorem 1.4.5 (a proof you have probably not seen) in UA. Try to outline the remainder of the proof of Theorem 1.4.5 in Exercise 1.4.7.
- Come to class with a draft of a proof from your Pick & Prove assignments.

Proof Structures

-
-
-
-
-
-
-

	$\forall x$	$\exists x$
to prove		
to use		

1. Assume that if A and B are countable, then $A \cup B$ is countable.

(a) Use induction to prove:

Theorem. *If $A_1, A_2, \dots, A_m$ are countable sets, then $A_1 \cup A_2 \cup \dots \cup A_m$ is countable.*

(b) Why can't we use induction for this one?

Theorem. *If A_n is countable for each $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} A_n$ is countable.*

Axiom of Countable Choice (uncontroversial)

Given a countable number of nonempty sets $A_1, A_2, \dots$, we can arbitrarily select a single element from each set, forming a corresponding sequence of elements $x_1, x_2, \dots$

Enumeration Principle: By definition, every countable set has (at least one) enumeration.

ACC: “we can pick one of these enumerations for each of countably many sets”

Proof Writing Guidelines

1. **Assume that the reader of the proof knows less than the writer.** Write as if your audience consists of fellow students and not simply the professor. Do not use the fact that the instructor already knows the answer as an excuse to be vague.
2. **There will always be differences in your scratch work/draft and final proof.** This is okay! Just make sure you're writing the final proof using these guidelines.
 - You may find it useful to work out a proof by working backwards, starting at the conclusion and getting to the hypothesis. However, your final proof should flow from beginning to end.
 - Make sure you're turning your *intuition* into an actual *formal proof*. Often (but not always), it'll get shorter!
3. **The structure of your proof should be clear.** The reader should be able to tell quickly what proof technique you're using, and what the different parts of your proof are. This isn't an essay — you're allowed to have headings, indentations, etc. instead of a whole block of text.
 - Assumptions and conclusions should always be included. Don't leave them out because they appear in the statement of the theorem.
 - Start your proof with *Proof.* and end your proof with $\square$, *QED*, or similar.
 - In a proof by cases, label each part with the claim you are proving.
 - “For the sake of contradiction assume...”, $(\Rightarrow)$, $(\Leftarrow)$
4. **Every step should clearly follow from the previous steps.** You aren't a textbook that's trying to leave things to the reader. Explain why things follow from other statements you've made.
5. **If you're stating what *needs* to be proved, make it clear that's what you're doing.** The reader should understand which statements are things you're supposing, and which statements you're stating in order to be able to prove them later.
 - “To prove $a < \frac{m}{n} < b$, it suffices to show $na < m < nb$.”
 - “Claim: The function f is onto.” This would be followed by a proof of this statement (as one of multiple claims that you needed to prove).
6. **If you're using a previously proved theorem, name it or explain why.** Don't leave the reader guessing about where your assumption comes from.
 - “... by the Axiom of Completeness...”
 - “... since the countable union of countable sets is countable.”
 - Avoid statements such as “by theorem 1.2.4,” since they are not universal.
7. **Be concise.** Sometimes this may seem at odds with the previous three guidelines. You should include exactly what you need, and nothing more. This comes with practice.
 - Don't include everything you know about the statement in case some of it is relevant and correct.
 - Don't include an example to illustrate your point.

Proof Writing Guidelines, cont.

8. **The proof should be written in complete sentences.** Remember, every mathematical symbol can be read as a word and the grammar should make sense when you read a sentence out loud. Sentences should start with capital letters and have periods at the end. Mathematical symbols should not start a sentence.
- “Choose $n \in \mathbb{N}$ such that $\frac{1}{n} < b - a$.” is a full sentence.
 - “ $n \in \mathbb{N}$, $\frac{1}{n} < b - a$.” is not a sentence. It’s not clear where n came from, and it needs connecting words between the two statements.
9. **Use mathematical symbols correctly, and don’t overdo it.** A few symbols, like π , can be used without further explanation. If you introduce your own notation, define it. Follow notational conventions (for example, don’t call a real number ξ unless you have a very good reason). Sentences shouldn’t be composed strictly of symbols, use English words where it makes things clearer.

Conventions for variable naming (in this class)

- Real numbers: $x, (y, z, w), a, (b, c, d)$
- Rational numbers: $r, (p, q)$
- Sets: $A, (B, C)$
- Functions: $f, (g, h)$
- Intervals: $I, (J)$
- Natural numbers: $n, (m)$
- Indices: $i, n, (j, k, m)$
- Large bounds: $N, (M)$
- *Never* name a variable o, e, ϵ , or δ .

Fun proof-y words and phrases you can use (correctly!)

- *thus, therefore, hence, so, because, since, ergo, consequently, it follows that*
- *consider a particular but arbitrarily chosen, let, suppose, assume*
- *which contradicts the supposition, now the absurdity is at hand, $\rightarrow\leftarrow$ (for a contradiction)*
- *without loss of generality assume* (when the assumption is not explicitly given but the particulars don’t restrict you. e.g. we have two different integers a and b , WLOGA $a < b$.)
- *by symmetry* (when there are technically multiple cases but the proofs play out exactly the same way)
- *indeed* (yes, that thing is what we thought it was, here’s why)
- Avoid: *trivially, clearly, obviously, of course*

After Class:

No problems today, finish your proofs for Friday.

Understanding Analysis Sections Covered: Sections 2.2-2.5

Definitions and Theorems to include in Analysis Journal:

- Definition 2.2.1 (sequence)
 - Definition 2.3.1 (bounded sequence)
 - Definition 2.4.1 (inc., dec., monotone)
 - Definition 2.5.1 (subsequence)
-

Before Class:

Readings: *How to Think about Analysis*, Chapter 5: Sequences, 5.1-5.4 (pp. 55-67)

Analysis Journal Prompts:

- Think about the 7 statements on p. 66 of Alcock and decide whether you think each is true or false. Try to provide a quick justification (not a proof) of the true ones and try to come up with a counterexample for the false ones.
- The statements on p. 67 are more challenging, but give them a shot as well.

True or False: Intuition and ideas

- Every bounded sequence is convergent.
- Every convergent sequence is bounded.
- Every monotonic sequence is convergent.
- Every convergent sequence is monotonic.

- Every monotonic sequence is bounded.
- Every bounded sequence is monotonic.
- Every bounded monotonic sequence is convergent.
- Every convergent sequence has a monotonic subsequence.
- Every sequence has a monotonic subsequence.
- Every bounded sequence has a convergent subsequence.

Convergence

A sequence (a_n) converges to a limit a if and only if...

Informal definition:

Formal definition:

After Class:

Pick & Prove Problems:

★ Exercise 2.2.4

Give an example of each of the following, or argue that such a request is impossible.

- (a) A sequence with an infinite number of ones that does not converge to one.
- (b) A sequence with an infinite number of ones that converges to a limit not equal to one.
- (c) A divergent sequence such that for every $n \in \mathbb{N}$ it is possible to find n consecutive ones somewhere in the sequence.

★ Exercise 2.2.7a,b,d

Use the definitions of *eventually* and *frequently* given in your book (Exercise 2.2.7).

- (a) Is the sequence $(-1)^n$ eventually or frequently in the set $\{1\}$?
- (b) Which definition is stronger? Does frequently imply eventually or does eventually imply frequently?
- (d) Suppose an infinite number of terms of a sequence (x_n) are equal to 2. Is (x_n) necessarily eventually in the interval $(1.9, 2.1)$? Is it frequently in $(1.9, 2.1)$?

★ Exercise 2.5.1 b-d

Give an example of each of the following, or argue that such a request is impossible.

- (b) A sequence that does not contain 0 or 1 as a term but contains subsequences converging to each of these values.
- (c) A sequence that contains subsequences converging to every point in the infinite set $\{1, 1/2, 1/3, 1/4, 1/5, \dots\}$.
- (d) A sequence that contains subsequences converging to every point in the infinite set $\{1, 1/2, 1/3, 1/4, 1/5, \dots\}$, and no subsequences converging to points outside of this set.

Understanding Analysis Sections Covered: Sections 2.2-2.3

Definitions and Theorems to include in Analysis Journal:

- Definition 2.2.3 (convergence of a sequence)
- Definition 2.2.3B (convergence of a sequence: topological version)
- Theorem 2.2.7 (uniqueness of limits)
- Theorem 2.3.2 (every convergent sequence is bounded)

Before Class:

Readings:

- *How to Think about Analysis*, Chapter 5: Sequences, 5.6-5.7 (pp. 71-75)
- *Understanding Analysis*, Discussion about Quantifiers and “Template for a Proof” on p. 45 (half a page)

Analysis Journal Prompts:

- Fully acquaint yourself with the proof structure outlined in the “Template for a Proof.” The definition of convergence has three quantifiers in it, so line up the definition of convergence with their proof template. Which statements/tasks come from which quantifiers?
- Exercise 2.2.3 (working with quantifiers): Describe what we would have to demonstrate in order to disprove the following statements.
 - (a) At every college in the US, there is a student who is at least 7 feet tall.
 - (b) For all colleges in the US, there exists a professor who gives every student a grade of either A or B.

Convergence: Formal Definition

(a_n) converges to a if and only if...

Other ways to write “ (a_n) converges to a ”:

1. What would we need to do to show that (a_n) *does not* converge to a ?

2. Show $\lim \left(\frac{n+1}{n} \right) = 1$.

- Step 1: Outline the proof.
- Step 2: Find N (work backwards).
- Step 3: Write a formal proof (forwards).

Convergence: Topological Definition

(a_n) converges to a if and only if...

3. Show that $(5, 4, 3, 2, 1, 0, 0, 0, 0, 0, \dots)$ converges to 0.

Theorem 2.3.2

Every convergent sequence is bounded.

Proof.

□

After Class:**Pick & Prove Problems:****★ Exercise 2.3.7ace**

Give an example of each of the following.

- (a) Sequences (x_n) and (y_n) which both diverge, but whose sum converges.
- (c) A convergent sequence (b_n) with $b_n \neq 0$ for all n such that $(1/b_n)$ diverges.
- (e) Two sequences (a_n) and (b_n) , where (a_nb_n) and (a_n) converge but (b_n) does not.

★★ Exercise 2.2.5

Let $[[x]]$ be the greatest integer less than or equal to x . For each sequence, find $\lim a_n$ and prove it using the definition of convergence.

$$(a) \ a_n = \left[\left[\frac{5}{n} \right] \right]$$

$$(b) \ a_n = \left[\left[\frac{12 + 4n}{3n} \right] \right]$$

★★ Exercise 2.2.6

Prove Theorem 2.2.7 (The limit of a sequence, when it exists, must be unique). To get started, assume $(a_n) \rightarrow a$ and also $(a_n) \rightarrow b$. Now argue $a = b$.

★★ Exercise 2.3.3 (Squeeze Theorem)

Show that if $x_n \leq y_n \leq z_n$ for all $n \in \mathbb{N}$, and if $\lim x_n = \lim z_n = l$, then $\lim y_n = l$ as well.

Understanding Analysis Sections Covered: Section 2.4

Definitions and Theorems to include in Analysis Journal:

- Theorem 2.4.2 (MCT)

Before Class:

Readings: None

Analysis Journal Prompts:

- Think about how you would use the definitions of monotone, bounded, and convergence to prove the Monotone Convergence Theorem.

Monotone Convergence Theorem

Every bounded, monotone sequence is convergent.

Scratch work:

Proof.

□

Recursive Sequences

Consider the following recursively defined sequence. Is it monotone? Bounded? Convergent? If so, what does it converge to? Prove your findings.

$$x_1 = 3, \quad x_{n+1} = \frac{1}{4 - x_n}$$

After Class:

Pick & Prove Problems:

★ Exercise 2.3.12+

A typical task in analysis is to decipher whether a property possessed by every term in a convergent sequence is necessarily inherited by the limit. Assume $(a_n) \rightarrow a$ and determine the validity of each claim. Try to produce a counterexample for any that are false (no justification is needed for those that are true).

- (a) If every a_n is an upper bound for a set B, then a is also an upper bound for B.
- (b) If every a_n is in the complement of $(0, 1)$, then a is in the complement of $(0, 1)$.
- (c) If every a_n is rational, then a is rational.
- (+) If every $a_n \in (0, 1)$, then $a \in (0, 1)$.

★ Exercise 2.4.2

- (a) Determine what is wrong with the following argument:

Consider the recursively defined sequence $y_1 = 1$, and $y_{n+1} = 3 - y_n$. Because y_n and y_{n+1} have the same limit, taking the limit across the recursive equation gives $y = 3 - y$. Solving for y , we conclude that $\lim y_n = 3/2$.

- (b) This time set $y_1 = 1$ and $y_{n+1} = 3 - \frac{1}{y_n}$. *Why does the strategy work this time?* Compute the limit of the sequence.

★ ★ ★ Exercise 2.4.4

- (a) In Section 1.4 we used the AoC to prove the Archimedean Property. Show that the Monotone Convergence Theorem can also be used to prove the Archimedean Property without making any use of AoC.
- (b) Use the Monotone Convergence Theorem to supply a proof for the Nested Interval Property that doesn't make use of AoC.

These two results suggest that we could have used the Monotone Convergence Theorem in place of AoC as our starting axiom for building a proper theory of the real numbers.

Understanding Analysis Sections Covered: Section 2.3

Definitions and Theorems to include in Analysis Journal:

- Theorem 2.3.3 (algebraic limit theorem)

Before Class:

Readings: *How to Think about Analysis*, Section 5.10: Combining Convergent Sequences (pp. 81-83)

Analysis Journal Prompts:

- In the proof in the reading, discuss why/how we're choosing ϵ carefully when we can.

Theorem 2.3.3: Algebraic Limit Theorem

Suppose $(a_n) \rightarrow a$ and $(b_n) \rightarrow b$. Then

$$(i) \ (c \cdot a_n) \rightarrow \qquad (ii) \ (a_n + b_n) \rightarrow \qquad (iii) \ (a_n \cdot b_n) \rightarrow \qquad (iv) \ \left(\frac{a_n}{b_n} \right) \rightarrow$$

Scratch work:

Proof.

□

After Class:

Pick & Prove Problems:

★ Supplemental Exercise

Provide a counterexample to prove the following theorem false: If $(a_n) \rightarrow a$, then $([a_n]) \rightarrow [a]$ (where $[a]$ is the greatest integer function described in Exercise 2.2.5).

★★ Exercise 2.3.1a

If $(x_n) \rightarrow 0$, show that $(\sqrt{x_n}) \rightarrow 0$. (You can assume $x_n \geq 0$ for all $n \in \mathbb{N}$.)

★★ Exercise 2.3.2b

Prove that if $(x_n) \rightarrow 2$, then $(1/x_n) \rightarrow 1/2$. (Read and follow the proof of part iv) of the Algebraic Limit Theorem in the book.)

Understanding Analysis Sections Covered: Sections 2.3, 2.5

Definitions and Theorems to include in Analysis Journal:

- Theorem 2.3.4 (order limit theorem)
- Exercise 2.4.7 (definition of $\limsup$ and $\liminf$)
- Theorem 2.5.2 (subsequences converge to same limit)

Before Class:

Readings: None

Analysis Journal Prompts:

- Think about how to prove the remaining part(s) of the Algebraic Limit Theorem. How should we go about choosing an epsilon that works within our proof?

Theorem 2.3.4: Order Limit Theorem

Suppose $(a_n) \rightarrow a$ and $(b_n) \rightarrow b$. Then

- (i) If $a_n \geq 0$ for all $n \in \mathbb{N}$, then
- (ii) If $a_n \leq b_n$ for all $n \in \mathbb{N}$, then
- (iii) Suppose there exists $c \in \mathbb{R}$ such that $c \leq b_n$ for all $n \in \mathbb{N}$. Then
- (iv) Suppose there exists $c \in \mathbb{R}$ such that $c \geq a_n$ for all $n \in \mathbb{N}$. Then

Idea: Use (i) to prove (ii), and use (ii) to prove (iii) and (iv).

Scratch work:

Proof.

□

Subsequences

Theorem 2.5.2

Subsequences of a convergent sequence converge to the same limit as the original sequence.

Proof.

□

1. How does this theorem give us an easy way to prove a sequence diverges?

2. Use the theorem to show $(b^n) \rightarrow 0$ for any $0 < b < 1$.

After Class:

Pick & Prove Problems:

★ Exercise 2.5.2abd

Decide whether the following propositions are true or false, providing a short justification for each conclusion.

- (a) If every proper subsequence of (x_n) converges, then (x_n) must converge as well.
- (b) If (x_n) contains a divergent subsequence, then (x_n) diverges as well.
- (d) If (x_n) is monotone and contains a convergent subsequence, then (x_n) converges as well.

★★ Exercise 2.5.6

Use a similar strategy to the one in Examples 2.5.3 to show that $\lim b^{1/n}$ exists for all $b \geq 0$ and find the value of the limit. (You may assume the results in Exercise 2.3.1.)

★★★ Exercise 2.4.7 (Limit Superior and Limit Inferior)

Let (a_n) be a bounded sequence.

- (a) Prove that the sequence defined by $y_n = \sup\{a_k : k \geq n\}$ converges.
- (b) The limit superior of (a_n) , or $\limsup a_n$, is defined by $\limsup a_n = \lim y_n$, where y_n is the sequence from part a). Provide a reasonable definition for $\liminf a_n$ and briefly explain why it also exists for any bounded sequence.
- (c) Prove that $\liminf a_n \leq \limsup a_n$, and give an example of a sequence for which the inequality is strict.
- (d) Show that $\liminf a_n = \limsup a_n$ if and only if $\lim a_n$ exists. In this case, all three share the same value.

Understanding Analysis Sections Covered: Sections 2.3, 2.5

Definitions and Theorems to include in Analysis Journal:

- Theorem 2.5.5 (Bolzano-Weierstrass)
 - Definition 3.2.5 (limit point)
-

Before Class:

Readings: None

Analysis Journal Prompts:

- Without reading the proof in the book, think about what might be involved in proving the Bolzano-Weierstrass Theorem stated below. How could we rewrite this statement formally using quantifiers, and how will that outline the structure of our proof? What definitions will we need to use?
- Let's get into some of these names: Bolzano and Weierstrass were some of the founding fathers of analysis. What other names can you list from Calculus/Differential Equations?

Theorem 2.5.5 (Bolzano-Weierstrass)

Every bounded sequence has a convergent subsequence.

Scratch work:

Proof.

□

Bolzano-Weierstrass Theorem (Topological Version)

Limit Point

A point x is a **limit point** (**cluster point/accumulation point**) of a set A if...

(1)

(2)

(3)

Are some of these definitions “stronger” than others? Why or why not?

After Class:

Pick & Prove Problems:

★ ★ ★ Exercise 2.5.4

The Bolzano-Weierstrass Theorem is extremely important, and so is the strategy employed in the proof. To gain some more experience with this technique, assume the Nested Interval Property is true and use it to provide a proof of the Axiom of Completeness. To prevent the argument from being circular, also assume that $(1/2^n) \rightarrow 0$. Explain why this last assumption is needed to avoid circularity.

★ ★ ★ Exercise 2.5.8

Another way to prove the Bolzano-Weierstrass Theorem is to show that every sequence contains a monotone subsequence. A useful device in this endeavor is the notion of a *peak term*. Given a sequence (x_n) , a particular term x_m is a peak term if no later term in the sequence exceeds it; i.e. if $x_m \geq x_n$ for all $n \geq m$.

- (a) Find examples of sequences with zero, one, and two peak terms. Find an example of a sequence with infinitely many peak terms that is not monotone.
- (b) Prove that every sequence contains a monotone subsequence and explain how this furnishes a new proof of the Bolzano-Weierstrass Theorem.

Understanding Analysis Sections Covered: Section 2.6

Definitions and Theorems to include in Analysis Journal:

- Definition 2.6.1 (Cauchy sequence)

Before Class:

Readings: None

Analysis Journal Prompts:

- Read and translate the definition of a Cauchy sequence below. What is it saying? What does it mean for a sequence to be Cauchy?

Cauchy Sequence

(a_n) is a **Cauchy** sequence if and only if

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n, m > N, |a_n - a_m| < \epsilon.$$

Are the following sequences Cauchy?

- $0, 1, 0, 1, 0, \dots$

- $\left(\frac{1}{n}\right)$

After Class:

Pick & Prove Problems:

★ **Exercise 2.6.2**

Give an example of each of the following, or argue that such a request is impossible.

- (a) A Cauchy sequence that is not monotone.
- (b) A Cauchy sequence with an unbounded subsequence.
- (c) A divergent monotone sequence with a Cauchy subsequence.
- (d) An unbounded sequence containing a subsequence that is Cauchy.

Understanding Analysis Sections Covered: Section 2.6, 8.2

Definitions and Theorems to include in Analysis Journal:

- Theorem 2.6.2
- Lemma 2.6.3
- Theorem 2.6.4 (Cauchy Criterion)
- Definition 8.2.2 (convergence in a metric space)
- Definition 8.2.3 (Cauchy sequence in a metric space)
- Definition 8.2.4 (completeness of a metric space)

Before Class:

Readings: None

Analysis Journal Prompts:

- How are Cauchy sequences related to convergent sequences? Must convergent sequences be Cauchy? Must Cauchy sequences be convergent? Justify your answers to yourself.

Theorem 2.6.2

Scratch work:

Proof.

Theorem 2.6.3

Scratch work:

Proof.

□

Theorem 2.6.4 (Cauchy Criterion)

Proof.

□

Why did we do this? Why do we need an equivalent definition of convergence that's much harder to work with?

After Class:

Pick & Prove Problems:

★ **Exercise 8.2.4**

Show that (in a metric space (X, d)), every convergent sequence is Cauchy.

★★ **Exercise 2.6.3a (modified)**

Prove that if (x_n) and (y_n) are Cauchy sequences, then $(x_n + y_n)$ is a Cauchy sequence. (You may not use the Cauchy criterion or Algebraic Limit Theorem.)

★★ **Exercise 8.2.3, 8.2.5a**

Consider $\mathbb{R}^2$ with the discrete metric $\rho(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$.

(a) Verify that this is actually a metric.

(b) What do Cauchy sequences look like in this space? Is $\mathbb{R}^2$ complete with respect to this metric?

★★ **Supplemental Exercise (for those of you taking Graph Theory)**

Consider an undirected connected graph of your choosing, and consider the set of vertices with the metric $d(u, v) =$ the length of the shortest path from u to v .

(a) Verify that this is actually a metric. What goes wrong if the graph is not connected? What goes wrong if it's a directed graph?

(b) What do Cauchy sequences look like in this space? Is this space complete with respect to this metric?

Before Class:

Readings: None

Analysis Journal Prompts:

- Finish your Definition and Theorem map for these first two chapters of the textbook. Which proofs rely on which theorems/definitions?
- Come to class with an updated “Questions about Analysis” list. We’ll try to answer some in review for the test.

